Lesson Aim

Explain wine sensory science and how consumers interact with wines.

INTRODUCTION

Special senses include the sensory modalities of smell, taste, hearing, vision and balance. These are all necessary if we are to notice specific alterations and changes in our environments. These sensations all play a role in deepening the impact of our surroundings, such as food, beverages, music, or nature.

Sensation can also have a great effect on our organ systems. Seeing a full buffet, especially when we are hungry, can start the whole digestive process. Hearing a beautiful song can relax our cardiovascular system by affecting the autonomic nervous system.

WINE SENSORY SCIENCE

Many different sources of wines are available on the wine marketplace, including different varietals (wines made using a single named grape variety), vintages (wines made from grapes picked in a specific year), blends (wines made by combining different varietals), producers, and regions.

Wine sensory science provides tools to measure differences between these different sources, by analysing and interpreting reactions to products based on smell, taste, and sight.

With regards to wine, the following must be evaluated:

- How do extrinsic factors, such as the weight of the bottle and label design, influence consumer behaviour?
- What expectations about taste do consumers have?
- Are these expectations met?
- How do consumers perceive the wines they purchase?

Sensory science can be used to extract this information from consumers implicitly without asking them directly.

This is important because most wine consumers cannot describe which aromas or textures they perceive when tasting wine. Wine tasting requires practice and knowledge of the characteristics one is looking for in order to accurately assess a wine. Non-expert consumers are often able to tell whether or not they like a wine but generally cannot explain exactly why they like it.

By determining which specific characteristics, a wine consumer likes or dislikes, wine makers or sellers can offer better service.

Determining Consumer Preferences

Product sensory assessment must be carried out it in a way that limits bias, to reduce errors generated by the evaluation proce
itself.

Each wine sample must be served “blind” without revealing its source. Each sample must be served in identical glasses and be poured in the same way, so that volatiles are collected equally. The order in which samples are provided to tasters across the panel must vary, so that the same wine is not always tasted first or last.

Common tests include:

- Hedonic testing - to determine trends in the preferences of a particular consumer group.
- Difference testing - to determine whether wines are significantly different.
- Preference testing - to determine whether consumers prefer one wine over another.
- Descriptive analysis - to determine wine aromas and flavours.

Testing methods include:

- Laboratory tastings - Consumers are brought into a research laboratory to perform tastings under controlled conditions.
- Free sorting - Consumers are presented different samples and are asked to evaluate these with regards to taste, smell, or both, and then sort them into groups according to similarity and difference based on their own perception.

Data Analysis

Data can be analysed and represented using multidimensional scaling or principal components analysis.

The information gained from this analysis can then be used by wine makers and sellers to create sensory profiles of different types of customers.

TYPES OF SENSES

Sense of Smell: Olfaction

The receptors for olfaction are known as bipolar neurons. These neurons are stored in the nasal epithelium with olfactory glands. Axons from olfactory receptors form the olfactory nerves, which relay messages to olfactory bulbs. These then transmit to the limbic system and temporal and frontal lobes of the cerebral cortex.

Sense of Taste: Gustation

The gustatory receptor cells are stored in the taste buds. Dissolved chemicals known as tastants stimulate gustatory receptors by binding to receptors attached to G-proteins in the membrane or passing through ion channels. *Gustatory receptor cells then produce receptor potentials, which cause the release of neurotransmitters.* This will then stimulate nerve impulses in first-order sensory neurons. Gustatory receptor cells trigger nerve impulses in cranial nerves VII, IX and I. Signals also pass to the medulla oblongata, thalamus, and cerebral cortex (parietal lobe).

Vision

Refraction of light on the cornea and lens reflects an image on the retina, which focuses an inverted image on the central fovea on the retina. Transduction of light energy into a receptor potential occurs in the outer segment of rods and cones. Receptor potentials in rods and cones decrease the release of inhibitory transmitters, which induces graded potentials in bipolar and horizontal cells. Horizontal cells transmit inhibitory signals to bipolar cells, bipolar and amacrine cells transmit excitatory signals to ganglion cells, which depolarize and initiate nerve impulses. Ganglion nerve impulses are then conveyed into the optic nerve II fibre to the thalamus and from there to the cerebral cortex.

WINE EVALUATION

Wines are evaluated using three senses: sight, smell, and taste.

Sight

Sight is evaluated according to both clarity and colour. Good wines should be clear, without any visible suspended solid particles. They are poured into a thin, clean, clear glass to examine. A wine taster should use a standard light source. Some winemakers prefer always to use a candle flame because the light intensity and other characteristics are close to the same wherever they go. The wine glass is normally held against a standardised white background.

Wines are then judged according to colour, cloudiness, and dullness. With age, characteristics do change. White wines can take on golden or amber tones and reds can gain a brownish tone. Small crystalline deposits can develop in old wines and brilliance can fade. Judges consider these aging factors when judging older wines.

Smell

Smell is sensed through a small nasal cavity called the olfactory bulb. Particles from the wine can both reach the olfactory bulb via the nose (carried by air entering via the nose) and from vapours rising from the back of the mouth up into the olfactory bulb. Vapours that reach the olfactory bulb via the back of the mouth are generally considered part of taste, while particles entering via the nose are considered as smell. Wines are swirled in a glass and sniffed for 30 to 40 seconds to judge smell. Smelling a wine for longer is considered to reduce accuracy through confusion of the senses.

Taste

Taste is generally judged in terms of five different characteristics:

- Acidity
- Sweetness
- Bitterness
- Saltiness
- Feel/Touch

Acidity

Taste buds on the sides of the tongue are those that sense sourness. Wines can contain different proportions of different types of aids, and that can impact the sharpness or intensity sensed. Malic and citric acids are fairly sharp. Lactic and tartaric acids taste softer. In cool climate grown grapes, the malic acid content can be higher in grapes that are picked earlier; and as a result, wines made from such grapes will be sharper in taste.

Sweetness

Sugars are detected mostly on the tip if the tongue.

Sweetness may be increased by two things:

- More unfermented sugar remaining in the wine
- Presence of less acid in the wine

Sweetness is a desirable characteristic in some wines but should never be overpowering.

Bitterness

Bitterness is best detected by taste buds at the back of the tongue. This is an astringent taste not to be confused with sourness which is sensed on the front palette rather than the back.

Bitterness is caused by tannins and phenolic compounds derived from fruit skins, seeds, stalks, and wood of wine casks. Over time tannins do change in wines to become chemicals that are less bitter. Red wines are typically more bitter than white wines; and older red wines less bitter (more mellow) than younger red wines.

Saltiness

Saltiness is detected by the mid palette of the tongue. It is a taste that is not common in wines but can occasionally occur in certain types of wines from certain localities and in wines that contain higher levels of metals (iron or copper).

Feel or Touch

Feel or touch is a sensation that is sometimes referred to in wine tasting. Strictly speaking it is not a taste. It refers to a feeling that the wine is thin or watery, through to feeling thicker or less watery. This feeling is generally due to a combination of alcohols, sugars, other compounds, and a level of astringency (or lack of).

The tingling sensation caused by CO_2 in sparkling wines is also part of feel or touch.

Other Factors

Different people can perceive wines in different ways, and at different times.

Tasting is one of the most complex activities our brains perform. It draws on multisensory inputs. To perceive flavour, our brains must balance the relative contributions from taste, smell, and the trigeminal nerve (a complex cranial nerve responsible for facial sensation and motor function, such as biting and chewing).

People's ability to taste varies as the number of taste buds that people have on their tongues varies. The more taste buds someone has, the better they can taste.

People's ability to smell also varies. On average, women have a better sense of smell than men.

A person's senses can be influenced temporarily and permanently by anatomical and physiological conditions. Allergies and sinus problems or illnesses can interfere with normal taste. Eyesight can become less reliable as a person gets older. Most adults require glasses by the time they are in their 40's.

People can also have different anosmia (losses of smell) affecting tasting judgements – some cannot perceive cork taints; others cannot perceive peppery notes, etc.

Temperature can affect taste. Cold temperatures enhance bitterness, while warm temperatures accentuate sweetness.

Temperature can also affect the wine's perceived texture. Cold temperatures cause a wine to feel lighter in the mouth.

WINE FOOD INTERACTION

Wine and food can interact. Wine-food interactions are based on the food and wine's structure, texture, and flavour.

Sweetness and acidity are structural components of great importance when matching food and wine. If the food served with the wine is sweet or sour, the perception of sweetness or sourness in the wine decreases.

Salt in food can affect the perception of astringency and acidity in wine. High salt levels in food can intensify the astringent effect tannins in red wines and increase the perception of acidity in high-acid white wines.

The alcohol in the wine, which provides a warm, tactile response, can intensify the hotness perceived in spicy foods.

Texture in food and wine generally refers to the mouthfeel it creates. A wine's body refers to its weight, i.e., how heavy it feels in the mouth and how long its flavours persist. Wines with higher concentrations of tannin and alcohol have more body. Generally, re wines have more body than white wines. Heavy-bodied wines are best paired with heavier, chewier, richer foods. Light wines, on the other hand, are best paired with lighter, more delicate foods.